Western Mathematics

TOPIC REVISION
QUESTIONS

Mathematics Standard 2

MS-A1 Formulae and Equations

Section I –Multiple Choice Questions (1 Mark each)

Section II – Longer Questions (Marks indicated beside questions)

Section I

Multiple Choice Questions (1 Mark each)

15 marks

1. If $C = \frac{5}{9}(F - 32)$ the value of C when $F = 212$ is:

(A) 86

(B) 1056

(C) 100

(D) 3.68

2. The value of $\frac{a}{b^2} - c$ when $a = 24$, $b = 2$ and $c = -10$ is:

(A) - 4

(B) 0

(C) 6

(D) 16

3. Given the relationship; $d = 2e^2 + k$, and that $d = 16$, when $e = 2.5$. What is the value of k?

(A) -9

(B) 2.5

(C) 3.5

(D) 13.5

4. You are given the formula $v^2 = u^2 + 2as$.

If u is always positive, which is the correct formula for u?

(A) $u = \sqrt{v^2 + 2as}$

(B) $u = \sqrt{v^2 - 2as}$

(C) $u = v - 2as$

(D) $u = \sqrt{2as - v^2}$

5. The cost per student (C) for a Mathematics excursion is given by the equation

$C = \frac{140+5n}{n}$ where n is the number of students going on the excursion.
If the number of students going on the excursion increases from 35 to 40, what change does this make to the cost per student (C)? (Answer to the nearest five cents)

(A) Each student pays fifty cents less.

(B) Each student pays fifty cents more.

(C) Each student pays twenty five cents less.

(D) There is no change in the cost per student.

6. Use the formula $D = \frac{yA}{y+12}$ to find the value of D, when $y = 6$ and $A = 3$

(A) $D = \frac{1}{2}$

(B) $D = 1$

(C) $D = 18$

(D) $D = 15$

7. The formula $s = ut + \frac{at^2}{2}$ is rearranged to make a the subject. The result is:

(A) $a = \frac{2s - ut}{t^2}$

(B) $a = \frac{2(s - ut)}{t^2}$

(C) $a = \frac{2s + 2ut}{t^2}$

(D) $a = t^2(2s - 2ut)$

8. Which is a solution to the equation $6x - \frac{3x}{2} = 9$

(A) $x = -2$

(B) $x = -1$

(C) $x = 1$

(D) $x = 2$

9. Harry is working as a nurse on a children's ward. He uses Young's formula to calculate an 8 year old child's dose of Adamine.

Young's Formula : Dosage for child 1-12 $= \dfrac{\text{age of child (in years)} \times \text{adult dose}}{\text{age of child (in years)} + 12}$

Given the adult dosage of Adamine is 15 mL, what is the child's dose?

(A) 1.5 mL

(B) 3 mL

(C) 6 mL

(D) 12 mL

10. Given the formula : $B = 2\pi\left(R + \frac{T}{2}\right) \times \frac{A}{360}$

What is the value of A, when $R = 20$, $T = 30$ and $B = 55$.

(A) $A = 0.5$

(B) $A = 5.0$

(C) $A = 45.0$

(D) $A = 90.0$

11. Mischa who is 8 years-old and weighs 52 kg, is prescribed a drug which has an adult dose of 39 mg.

Clark's formula is used for finding children's dosages.: Dosage $= \frac{\text{weight in kg} \times \text{adult dosage}}{70}$,

What is the correct dose for Mischa?

(A) 4 mg

(B) 29 mg

(C) 36 mg

(D) 52 mg

12. The table was completed using the rule $y = -3x + 4$.

x	-1	0	1
y	7	4	?

Find the missing value.

(A) -1

(B) 0

(C) 1

(D) 10

13. When $\frac{a-4}{3} = 4$, find the value of a.

(A) 8

(B) 12

(C) 16

(D) 20

14. The formula to convert temperatures between Celsius and Farenhiet is $C = \frac{5}{9}(F - 32)$.

When the temperature is 95° Farenheit, what is the temperature in Celsius?

(A) 35°

(B) 71°

(C) 85°

(D) 139°

15. What is the solution to the equation below?

$$4m - 5 = m + 16 \text{ ?}$$

(A) $m = 3\frac{2}{3}$

(B) $m = 4$

(C) $m = 7$

(D) $m = 10\frac{1}{3}$

Western Mathematics
Mathematics Standard 2
MS-A1 Formulae and Equations
REVISION QUESTIONS - Section II

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 16 (2 marks) **Marks**

Solve $9x - 7 = x + 5$. **2**

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Question 17 (2 marks)

Given that $x = 6$, $y = -3$ and $z = -12$, find the value of the following expressions.

(a) $x(y+z)$ **1**

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(b) $\dfrac{x-z}{y}$ **1**

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Question 18 (2 marks)

Use the formula $s = \frac{n}{2}(a + l)$ to find the value of a when $l = 36$, $n = 6$ and $s = 99$. **2**

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Question 19 (2 marks)

Evaluate the following expressions if $m = -6$, $n = 3$, $x = 2$ and $y = 4$.

(a) $12x^2y^3$. **1**

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(b) $\frac{12m^3}{3n^2}$. **1**

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Question 20 (4 marks)

Solve the equations:

(a) $\frac{2x-1}{3} + \frac{7-x}{2} = 4$ **2**

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(b) $5x + 3(2 - x) = \frac{2x}{3} + 12$ **2**

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Question 21 (2 marks)

Calculate a Blood Alcohol Content Estimate for a woman weighing 65 kg, who has had four standard drinks over two and a half hours (answer to 2 decimal places). **2**
Use the formula $BAC_{\text{Female}} = \frac{10N - 7.5H}{5.5M}$

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Question 22 (2 marks)

Make r the subject of the formula: **2**

$$P = \frac{29r}{7} + 2w$$

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Question 23 (2 marks)

A one-and-a-half-year-old child needs medicine for a fever. **2**

The adult dosage of the medicine is 40 ml.

Fried's formula: $\text{Dosage for children } 1 - 2 \text{ years} = \frac{\text{age (in months)} \times \text{adult dosage}}{150}$

Use Fried's formula to calculate the dosage for the child to the nearest ml.

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Question 24 (5 marks)

When on highway trips, Hannah drives her car at an average speed of 96 km/h.

(a) How long would a highway trip of 168 km take, in hours and minutes? **2**

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(b) A highway trip lasts for 2 hours and 15 minutes. How far did she travel? **1**

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(c) On a trip on a motorway she travels at a faster speed. She travelled 35 km in twenty minutes. What was her average speed for this trip? **2**

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Question 25 (6 marks)

The braking distance for a car, is the distance that the car travels from the point where the brakes are applied.

The braking distance (d m) for two different cars are given by the formulae below, where u is the speed at which the car is travelling in m/s.

$$\text{Car } M: \quad d = \frac{u^2}{21.6}$$
$$\text{Car } H: \quad d = \frac{u^2}{18.6}$$

(a) Convert 60 km/h to m/s, correct to 3 significant figures. **1**

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(b) If both cars were travelling at 60 km/h, what is the difference in their braking distances. **2**

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(c) The reaction distance is the distance that a car travels in the time that it takes for the driver to react and apply the brakes. **2**
Brad is a driver with a reaction time of 1.25 seconds.
At 60 km/h, how far would the car travel in this time?

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(d) The stopping distance is the braking distance, plus the reaction distance. **1**
What is the stopping distance for Car H, with Brad driving?

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Question 26 (4 marks)

Rick is a 24 year-old-male with a mass of 95 kg who consumes 5 standard drinks over a period of 3 hours.

(a) Use the formula below to estimate his BAC after his last drink (correct to 2 s.f.). **2**

$$BAC_{\text{Male}} = \frac{10N - 7.5H}{6.8M}$$

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(b) The formula : **Time**= $\boldsymbol{\frac{BAC}{0.015}}$ **;** gives the number of hours required for a person to stop consuming alcohol in order to reach zero *BAC*. **2**
How many hours and minutes would Rick have to wait for his BAC to reach zero (to the nearest minute)?

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End of Revision

Western Mathematics MS-A1 Formulae and Equations Revision

Section I - Worked Solutions

No	Working	Answer
1	$C = \frac{5}{9}(F - 32)$ $= \frac{5}{9}(212 - 32)$ $= 100$	C
2	$\frac{a}{b^2} - c = \frac{24}{2^2} - \ - 10$ $= 16$	D
3	$d = 2e^2 + k$ $16 = 2(2.5)^2 + k$ $16 = 12.5 + k$ $k = 16 - 12.5$ $k = 3.5$	C
4	$v^2 = u^2 + 2as$ $v^2 - 2as = u^2$ $u = \sqrt{v^2 - 2as}$	B
5	$C = \frac{140 + 5n}{n}$ $n = 35$ $C = \frac{140 + 5 \times 35}{35} = \$9$ $n = 40$ $C = \frac{140 + 5 \times 40}{40} = \$8.50$ Each student pays $50c$ less	A
6	$D = \frac{yA}{y + 12}$ when $y = 6$ and A $= 3$ $D = \frac{6 \times 3}{6 + 12} = \frac{18}{18} = 1$	B

7	$s = ut + \frac{at^2}{2}$ $s - ut = \frac{at^2}{2}$ $at^2 = 2(s - ut)$ $a = \frac{2(s - ut)}{t^2}$	**B**
8	$6x - \frac{3x}{2} = 9$ $12x - 3x = 18$ [$\times by$ 2] $9x = 18$ [Simplify LHS] $x = \frac{18}{9} = 2$ [divide by 2]	**D**
9	Dosage for child 1-12 $= \frac{\text{age of child (in years)} \times \text{adult dose}}{\text{age of child (in years)} + 12}$ $= \frac{8 \times 15}{8 + 12}$ $= \frac{120}{20}$ $=$ 6 mL	**C**
10	$B = 2\pi\left(R + \frac{T}{2}\right) \times \frac{A}{360}$ $55 = 2\pi\left(20 + \frac{30}{2}\right) \times \frac{A}{360}$ $55 = 2\pi(35) \times \frac{A}{360}$ $55 = 220 \times \frac{A}{360}$ $A = 55 \times \frac{360}{220}$ $A = 90$	**D**
11	Dosage $= \frac{52 \times 39}{70} = 28.97142857 = 29$ mg	**B**
12	$y = -3x + 4$ $y = -3(1) + 4 = 1$	**C**

13	$\frac{a-4}{3}=4$ $a-4=4\times 3$ $a-4=12$ $a=12+4=16$	**C**
14	$C=\frac{5}{9}(F-32)$ $C=\frac{5}{9}(95-32)$ $=35$	**A**
15	$4m-5=m+16$ $3m-5=16$ $3m=21$ $m=\frac{21}{3}\ =\ 7$	**C**

	MS-A1 Formulae and Equations Section II Solution	Marks
Question 16		
(a)	$9x - 7 = x + 5$ $8x - 7 = 5$ $8x = 12$ $x = \frac{12}{8} = 1.5$	2
Question 17		
(a)	$x(y+z) = 6(-3+-12)$ $= 6(-15)$ $= -90$	
(b)	$\frac{x-z}{y} = \frac{6-(-12)}{-3}$ $= \frac{18}{-3}$ $= -6$	
Question 18		
	$s = \frac{n}{2}(a + l)$ $99 = \frac{6}{2}(a + 36)$ $99 = 3(a + 36)$ $99 = 3a + 108$ $3a = -9$ $a = -3$	2
Question 19		
(a)	$12x^2y^3 = 12(2)^2(4)^3 = 3072$	1
(b)	$\frac{12m^3}{3n^2} = \frac{12(-6)^3}{3(3)^2} = -96$	1

Question 20		
(a)	$\frac{2x-1}{3}+\frac{7-x}{2}=4$ $\frac{6(2x-1)}{3}+\frac{6(7-x)}{2}=24$ $2(2x-1)+3(7-x)=24$ $4x-2+21-3x=24$ $x+19=24$ $x=5$	2
(b)	$5x+3(2-x)=\frac{2x}{3}+12$ $5x+6-3x=\frac{2x}{3}+12$ $2x+6=\frac{2x}{3}+12$ $2x=\frac{2x}{3}+6$ $6x=2x+18$ $4x=18$ $x=4.5$	2
Question 21		
(a)	$BAC=\frac{10N-7.5H}{5.5M}$ $=\frac{10(4)-7.5(2.5)}{5.5(65)}$ ≈ 0.05944 ≈ 0.06 (2 d.p.)	2
Question 22		
	$P=\frac{29r}{7}+2w$ $P-2w=\frac{29r}{7}$ $29r=7P-14w$ $r=\frac{7P-14w}{29}$	2

Question 23		
	$1\frac{1}{2}$ years $=$ 18 months Dosage $= \dfrac{18 \times 40}{150} = 4.8$ $=$ 5 ml to nearest ml	2
Question 24		
(a)	$S = \dfrac{D}{T}$ $96 = \dfrac{168}{T}$ $T = \dfrac{168}{96} = 1.75$ $=$ 1 hour and 45 min 2hours and 15 min $=$ 2.25 hours	2
(b)	$S = \dfrac{D}{T}$ $96 = \dfrac{D}{2.25}$ $D = 96 \times 2.25 = 216$ km	1
(c)	Twenty min $= \dfrac{20}{60} = \dfrac{1}{3}$ of an hour $S = \dfrac{D}{T}$ $S = \dfrac{35}{\frac{1}{3}} = 105$ km /h	2

Question 25		
(a)	$60 \text{ km/h} = 60000 \text{ m/h}$ $= \dfrac{60000}{60 \times 60} \text{ m/s}$ $= 16.666 = 16.7 \text{ m/s}$	1
(b)	Car M : $d = \dfrac{(16.7)^2}{21.6}$ $= 13 \text{ m}$ Car H : $d = \dfrac{(16.7)^2}{18.6}$ $= 15 \text{ m}$ Car H has a 2 m longer braking distance	2
(c)	$S = \dfrac{D}{T}$ $D = S\,T$ Distance $= 16.7 \times 1.25 = 20.875$ $= 21\,m$	2
(d)	Stopping distance $= 21 + 15 = 36\,m$	1

Question 26		
(a)	$BAC_{MALE} = \dfrac{10N - 7.5H}{6.8M}$ $= \dfrac{10 \times 5 - 7.5 \times 3}{6.8 \times 95}$ $= 0.04256966$ $= 0.043$ (2 s f)	2
(b)	Time $= \dfrac{BAC}{0.015}$ $= \dfrac{0.04256966}{0.015}$ $= 2.83797733$ $=$ 2 hrs and 50 min (nearest minute)	1